

Maciej Krzysztof Zuziak

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PROFILE

AI researcher specialising in the security and privacy of decentralised machine learning – federated learning, membership inference attacks, differential privacy, and data leakage. Marie Skłodowska-Curie Doctoral Fellow (under 10% acceptance rate) and Max Weber Fellow (under 3% acceptance rate) with 5+ years of experience delivering EU-funded research (€20M+ programmes) across academia and industry. Combines rigorous ML engineering with formal legal training in data rights and compliance – able both to design privacy-preserving systems and to assess their regulatory implications. Experienced in independent research leadership, interdisciplinary collaboration, and translating theory into deployable decentralised systems.

EXPERIENCE

- Institute of Science Tokyo** Oct 2026 – Dec 2026
Visiting Researcher (Incoming), School of Computing Tokyo, Japan
 - Invited research stay hosted by Jun Sakuma, continuing collaboration on backdoor propagation through the model supply chain.
- European University Institute (EUI)** Sep 2026 – Sep 2027
Max Weber Fellow, Department of Law Florence, Italy
 - Research fellowship awarded for independent research on **GDPR compliance and privacy considerations regarding Large Language Models**.
 - Selected as **one of 56 distinguished researchers** (out of more than 2000 applications) to conduct independent research under the supervision of senior researchers from the EUI Department of Law.
- University of Leeds** Jan 2026 – Jun 2026
Research Fellow, School of Computer Science Leeds, United Kingdom
 - Led independent research** on decentralised systems for Smart Grids, UAV coordination, and Digital Democracy, identifying new methodological directions for the lab.
 - Co-author on *The Optimization Trilemma* (IEEE ACSOS 2026; arXiv:2607.17311), extending decentralised multi-agent coordination to optimise system efficiency, individual comfort and fairness jointly.
 - Surveyed emerging research areas and co-authored **joint grant proposals** targeting Horizon Europe, UKRI, and ARIA funding schemes.
 - Mentored junior researchers and coordinated lab-critical projects, contributing to research continuity and delivery.
- National Research Council of Italy (CNR)** Nov 2021 – Dec 2025
Research Associate, Knowledge Discovery Lab Pisa, Italy
 - Core contributor to two Horizon-scale EU projects** (€11M+), focusing on federated personalisation, explainability, and privacy risk mitigation.
 - Published 6 peer-reviewed outputs** and delivered 1 journal article currently under review.
 - Designed and implemented reusable ML libraries** for decentralised learning, including controlled non-IID splits, clustered personalisation, and Shapley-based contribution analysis.
 - Contributed methodological components adopted across multiple project work packages and experimental pipelines.
 - Collaborated with leading European research institutions CNRS, Scuola Normale Superiore, Scuola Superiore Sant'Anna, and other European partners on tractable federated algorithms and data governance frameworks.
- Tellu AS** Sep 2023 – Sep 2024 (dual appointment)
Visiting Research Engineer, DevOps Department Oslo, Norway
 - Co-developed a production-grade last-mile IoT gateway** with autonomous onboarding and decentralised fleet and device-health management.
 - Designed decentralised Digital Twin architectures for device enrolment, metrics collection, and lifecycle monitoring.
 - Delivered technical documentation for **three Horizon Europe mid-term reviews, all positively evaluated**.
- Permanent Mission of the Republic of Poland to the United Nations** Jul 2019 – Aug 2019
Intern, Security Council New York, USA
 - Attended Security Council sessions on peacekeeping, counter-terrorism, and transnational crime; engaged with Sanctions Committees (DR Congo, Libya) and **commended by the Head of Mission**.

EDUCATION

• University of Pisa

Nov 2021 – Dec 2025

Ph.D. (Artificial Intelligence)

Pisa, Italy

- Thesis: *Data Collaboration with the Use of Federated Learning: Issues of Marginal Contribution Quantification and Personalisation* (evaluation: eccellente - highest distinction). - [Thesis Repository](#).
- Joint National Doctorate in AI coordinated with, among others, Scuola Superiore Sant'Anna, Scuola Normale Superiore, University of Trento, and University of Bologna.
- 300+ hours of advanced CS & mathematics modules (graded A–A+).
- All three annual appraisals rated exceptional.

• University of Silesia in Katowice

Sep 2016 – Jul 2021

Magister Prawa (Master of Laws, integrated five-year programme)

Katowice, Poland

- Thesis: *Digitalisation of Factoring Contracts* (evaluation: bardzo dobry - highest distinction).
- Post-graduation prize for most promising graduates.

JOURNALS

- [1] [\[First\]](#) M.K. Zuziak, R. Pellungrini, S. Rinzivillo (2025): **One-Shot Clustering for Federated Learning Under Clustering-Agnostic Assumption**. ACM Transactions on Intelligent Systems and Technology (revision under review) - [ArXiv Preprint](#).

PUBLICATIONS

FIRST = FIRST AUTHOR; CO-FIRST = EQUAL CONTRIBUTION, AS MARKED IN THE PUBLICATION

- [1] [\[Co-first\]](#) S.Z. El Mestari*, M.K. Zuziak*, G. Lenzini (2026): **Poison to Detect: Detection of Targeted Overfitting in Federated Learning**. *Proceedings on Privacy Enhancing Technologies* 2026(4), pp. 480–506. Presented at *The 26th Privacy Enhancing Technologies Symposium*, Calgary, Canada - [PoPETs Access](#). (*equal contribution)
- [2] J. Nikolic, M.K. Zuziak, E. Pournaras (2026): **The Optimization Trilemma: Efficiency, Comfort and Fairness in Decentralized Multi-agent Coordination**. At *IEEE ACSOS 2026 — 7th IEEE International Conference on Autonomous Computing and Self-Organizing Systems*, Cesena, Italy, 7–11 September 2026. IEEE (to appear in IEEE Xplore) - [ArXiv Preprint](#).
- [3] [\[Co-first\]](#) S.Z. El Mestari*, M.K. Zuziak*, G. Lenzini, S. Rinzivillo (2025): **Can Contributing More Put You at a Higher Leakage Risk? The Relationship Between Shapley Value and Training Data Leakage Risks in Federated Learning**. At *SECRYPT 2025*, Bilbao, Spain. SciTePress - [SciTePress Access](#). (*equal contribution)
- [4] [\[First\]](#) M.K. Zuziak, R. Pellungrini, S. Rinzivillo (2024): **One-Shot Clustering for Federated Learning**. At *IEEE Big Data 2024*, Washington, USA. IEEE Xplore - [IEEE Xplore Access](#).
- [5] [\[First\]](#) M.K. Zuziak, S. Rinzivillo (2024): **Amplified Contribution Analysis for Federated Learning**. At *IDA 2024*, Stockholm, Sweden. Springer Lecture Notes in Computer Science (LNCS) - [Springer Nature Access](#).
- [6] [\[Co-first\]](#) M.K. Zuziak*, O. Hinrichs*, A. Abdrassulova, S. Rinzivillo (2023): **Data Collaboratives with the Use of Decentralised Learning**. At *ACM FAccT 2023*, Chicago, USA. ACM - [ACM Digital Library Access](#). (*equal contribution)

POLICY CONTRIBUTIONS

- [1] ACM ETPC (2026). **Comments in Response to European Commission Call for Evidence Survey on the “Digital Omnibus on AI Regulation” Proposal**. Contributors: G. Besanson, M.K. Zuziak, A. Saucedo, F. Medeiros, T. Romanoff. [ACM ETPC Link](#).
- [2] ACM ETPC (2025). **Consultation on the Guidelines on Protection of Minors Online under the Digital Services Act**. Contributors: T. Romanoff, A. Saucedo, M.K. Zuziak, M. Beaudouin-Lafon, C.L. Hankin. [ACM ETPC Link](#).
- [3] ACM ETPC (2024). **Comments on Draft Delegated Regulation on Data Access (DSA)**. Contributors: A. Deshpande, L. DiBiasi, M.K. Zuziak, J. Rozanec, A. Nagy, A. Saucedo, F. Medeiros, J. Mizeria-Petraszko. [ACM ETPC Link](#).

WORKSHOPS, POSTERS & OPINIONS

W=WORKSHOPS, P=POSTERS, O=OPINIONS

- [P.3] S.Z. El Mestari*, M.K. Zuziak*, G. Lenzini (2026): **Poison to Detect: Detection of Targeted Overfitting in Federated Learning (Poster Version)**. Privacy Enhancing Technologies Symposium, Canada. (*equal contribution).
- [P.2] M.K. Zuziak, S. Rinzivillo (2024): **Personalization via Clustering in Federated Learning for Accessible Digital Services**. School of Advanced Studies Sant'Anna at Pisa Internet Festival, Italy.
- [O.1] M.K. Zuziak (2024): **How to Collaboratively Use Statistical Models in a Secure Way**. *Opinio Juris in Comparatione*. [Opinio Juris Link](#).

- [P.1] M.K. Zuziak, S. Rinzivillo (2022): **Federated Learning for Personal Data Management under European Digital Strategy**. Conference on Data Sciences and EU Regulations, IRIT at Université Toulouse III, France.
- [W.1] M.K. Zuziak, S. Rinzivillo (2022): **Federated Learning as an Analytical Framework for Personal Data Management - a Proposition Paper**. Hybrid Human Artificial Intelligence (HHAI) Workshop, Amsterdam, Netherlands. CEUR Proceedings - [CEUR WS Link](#).

RECENT TALKS AND SEMINARS

- [1] Institute of Science Tokyo (2026): **On the Intrinsic Vulnerabilities in Federated Learning and Corresponding Defence Methods**.
- [2] ACM (2026): **Europe's Digital Omnibus: A New Digital Regime or Simplification**. [ACM Recording](#).

SELECTED PROJECTS

- **Project Tango (Horizon Europe No. 101120763)** 2024 – 2025
Research Associate  [Project page](#)
 - **Total Funding:** €7M
 - **Key Partners:** Université Paris Cité, TU Darmstadt, University of Pisa, Scuola Normale Superiore.
 - **Role:** Research on clustered federated learning and explainability for human-centric decision-making.
- **Project LEADS (MSCA ITN No. 956562)** 2021 – 2024
Marie Skłodowska-Curie Actions (MSCA) Doctoral Fellow  [Project page](#)
 - **Total Funding:** €3.8M
 - **Key Partners:** University of Pisa, Scuola Superiore Sant'Anna, CNRS, University of Luxembourg.
 - **Role:** Research on trustworthy and decentralised ML infrastructure for the Digital European Strategy.
- **Project ENTRUST (Horizon Europe No. 101095634)** 2023 – 2024
Visiting Contributing Researcher  [Project page](#)
 - **Total Funding:** €5.6M
 - **Key Partners:** Tellu AS, TU Eindhoven, Siemens, Sintef.
 - **Role:** Contributed to decentralised trust management architecture; designed risk assessment processes for critical IoT infrastructure (last-mile medical gateways).
- **Project Eratosthenes (Horizon 2020 No. 101020416)** 2023 – 2024
Visiting Contributing Researcher  [Project page](#)
 - **Total Funding:** €6M
 - **Key Partners:** Tellu AS, TU Graz, KU Leuven, Airbus Cybersecurity, Sintef.
 - **Role:** Contributed to an autonomous onboarding flow for critical IoT infrastructure (last-mile medical gateways and sensors connected to them).

OPEN SOURCE

-  **FedJust** 2021–2024
Federated Learning toolkit for rapid prototyping and reproducible evaluation of novel federated algorithms. [GitHub](#)
-  **OCFL Suite** 2024–2025
Experimental framework for designing and benchmarking Federated Clustered Learning methods. [GitHub](#)

AWARDS

- **Max Weber Fellowship** 2026
European University Institute (under 3% acceptance rate)
- **Horizon Europe (Project Tango) Fellowship** 2024
National Research Council of Italy (CNR)
- **Marie Skłodowska-Curie Actions Fellowship** 2021
European Commission (under 10% acceptance rate)
- **CSC Scholarship** 2020
China Scholarship Council
- **Academic Scholarship (Top 5%)** 2017
University of Silesia (awarded multiple times 2017–2020)

MEMBERSHIPS

- **ACM Europe Technology Policy Committee (ETPC)**

Dec 2023 – Present

- Non-voting member

RESEARCH INTERESTS

- Federated and decentralised learning
- Privacy-preserving ML (differential privacy, membership inference, data leakage)
- LLM security: backdoor implantation and propagation through fine-tuning and distillation, data poisoning, jail-breaking
- Explainable AI and contribution analysis in distributed settings
- AI governance, liability, and regulation

SKILLS

- **Core Expertise:** Federated Learning, Privacy-Preserving ML, Explainable AI, Data Governance
- **Programming:** Python, Julia, C/C++, TypeScript
- **ML & Data:** PyTorch, TensorFlow/Keras, JAX, Scikit-Learn, NumPy, Pandas
- **Distributed & Systems:** Flower, PySyft/PyGrid, TensorFlow Federated, Docker, Kubernetes, Linux/WSL
- **Agentic & LLM Systems:** LangChain, LangGraph, RAG pipelines, vector databases (FAISS, Pinecone, Weaviate), LoRA/PEFT

ADDITIONAL INFORMATION

Languages: Polish (Native), English (C2), Chinese (B2), Italian (B1/B2)